

*Environmental Risk and Strategic Responsibility: A Game Theoretic Approach to Damage
Allocation in Pipeline Systems*

Introduction and Background

Across North America, pipeline systems have consistently failed to meet public safety expectations. Incidents such as the Yellowstone River oil spill (Inside Climate News, 2015) and the major leaks from the Keystone pipeline (The Guardian, 2022) reveal not only technical vulnerabilities but also institutional failures in the allocation and oversight of environmental risk. Furthermore, landowners often cannot observe regulatory enforcement, which undermines trust and reduces the incentive for precautionary action (Reuters, 2015; Kansas Reflector, 2022).

This challenge can be addressed through a game theoretic framework that captures the strategic behavior of stakeholders operating under uncertainty and conflicting incentives. Araujo and Leoneti (2018) use two by two strategic games to demonstrate how regulatory contexts influence cooperation in the oil and gas sector, although their model is limited to symmetric, two actor interactions. Eid, El Adaway, and Coatney (2015) provide a more dynamic and multi actor perspective through an evolutionary game model of post disaster insurance, illustrating how adaptive behavior and risk preferences contribute to system resilience. Building on these contributions, this study develops a formal model featuring three strategic actors, endogenous risk, partial information asymmetry, and heterogeneous utility structures.

The model examines how a liability sharing mechanism can be structured to induce responsible behavior, even in the presence of imperfect information and uncertainty. It shows that aligning expected damage with levels of precaution can lead to voluntary compliance and offers a framework for regulatory design in high risk, low transparency infrastructure systems.

Model

In this model, a pipeline leak occurs with probability q , determined jointly by nature and the strategic choices of the players. The resulting environmental damage, denoted M , is allocated among the pipeline owner, the state, and the landowner according to their respective levels of care, captured through behavioral weights. The pipeline owner (P) chooses between an action representing regulatory compliance (a) with a cost C_P , and an action representing cost cutting noncompliance (b). The state (S) chooses between active monitoring (c) with a cost C_S , or leniency (d). The landowner (L) chooses between precautionary behavior (e) with a cost C_L , and negligence (f). This is represented in the following table:

Players	Strategy	Action	Cost of Action
Pipeline Owner	a	Compliance	C_P
Pipeline Owner	b	Noncompliance	0
State	c	Monitoring	C_S
State	d	Leniency	0
Landowner	e	Caution	C_L
Landowner	f	Negligence	0

This damage sharing structure internalizes strategic responsibility rather than prescribing literal financial transfers. The state's share reflects the reputational, legal, or political consequences of inadequate oversight rather than direct monetary losses. Similarly, the shares borne by the pipeline owner and the landowner represent the internalized consequences of their decisions.

The game is structured as a sequential extensive form interaction with imperfect information. The pipeline owner moves first, followed by the state regulator who observes this action. The landowner, unable to observe the state's preceding moves, selects a strategy simultaneously. Each player is assigned a distinct utility function reflecting their respective risk preference.

Risk preferences are assigned based on each player's underlying incentives: the pipeline owner is modeled as risk seeking, the state as risk neutral, and the landowner as risk averse. These behaviors reflect their institutional roles—pipeline owners are incentivized to take risks for higher returns, states are expected to weigh tradeoffs without direct exposure to personal loss, and landowners, facing the most immediate and personal consequences of environmental harm, tend to be more loss averse. Environmental damage is distributed according to a weight scheme that assigns greater losses to those who act negligently, thereby internalizing external costs and encouraging responsible behavior. Because total damage is endogenously determined by collective actions along with the stochastic factor, each player's payoff depends not only on their own strategy but also on the choices of others, capturing the interdependent nature of shared responsibility here.

Therefore, the probability of a leak (q) increases from a baseline of $q_0 = 0.2$ (probability of a leak due to random natural causes) depending on the players' choices: it rises by 0.2 (100%) if the pipeline owner acts negligently, by 0.1 (50%) if the state regulator is lenient, and by 0.05 (25%) if the landowner fails to take precautions, reflecting each actor's relative control over pipeline safety. Formally, this is expressed as:

$$q = 0.2 + 0.2 \times 1 (\text{pipeline owner's action}) + 0.1 \times 1 (\text{state's action}) + 0.05 \times 1 (\text{landowner's action}),$$

where the indicator function equals 1 if a player chooses a negligent action and 0 otherwise. Furthermore, the responsible actions (a, c, and e) lead to damage sharing weight = 1 and the

negligent actions (b, d, f) lead to weight = 2 where W (Total Weight) = W_P (weight for pipeline owner) + W_S (weight for state) + W_L (weight for landowner) and the utility functions are:

Pipeline owner: $P = \frac{1}{10000} \left(\pi - C_P - qM \frac{W_{P=a}}{W} \right)^2$ if P plays a, $P = \left(\pi - qM \frac{W_{P=b}}{W} \right)^2$ if P plays b,

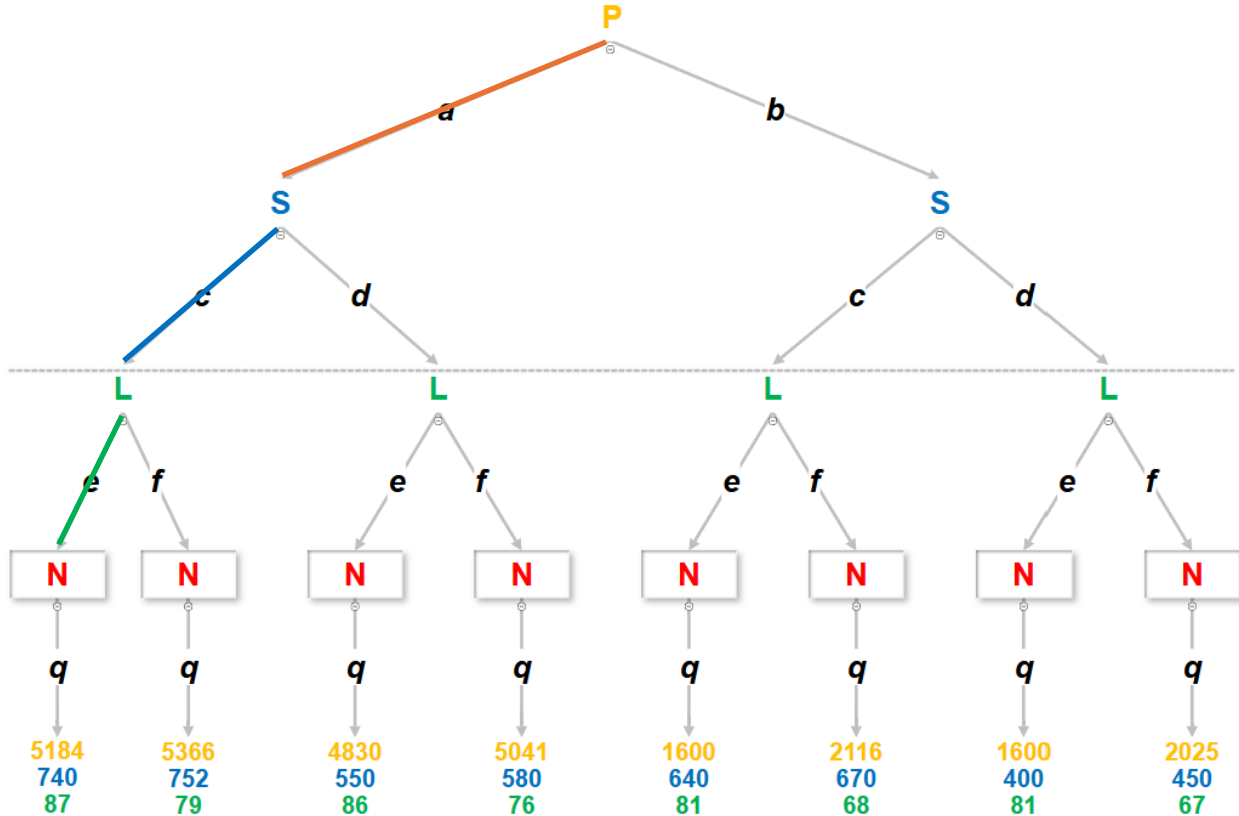
State: $S = \frac{1}{10} \left(\sigma - C_S - qM \frac{W_{S=c}}{W} \right)$ if S plays c, $S = \left(\sigma - qM \frac{W_{S=d}}{W} \right)$ if S plays d,

Landowner: $L = \left(\sqrt{\lambda - C_L - qM \frac{W_{L=e}}{W}} \right)$ if L plays e, $L = \left(\sqrt{\lambda - qM \frac{W_{L=f}}{W}} \right)$ if L plays f,

where π , σ , and λ are the initial wealth (endowment) of the pipeline owner, the state, and the landowner respectively.

Results and Discussion

The following payoffs are derived in the game by substituting the weights for each strategy and plugging in the values $\pi = \sigma = \lambda = 10,000$; $C_P = 800$; $C_S = 600$; $C_L = 400$; $M = 30,000$.



The distribution of costs reflects the relative burden of responsible actions in pipeline development: pipeline owners incur substantial compliance costs during the installment, the state bears moderate monitoring expenses, and landowners face relatively minor precautionary efforts. Additionally, equal initial endowments are assumed to isolate the effects of strategic behavior from differences in wealth. Due to asymmetric information, the state observes the pipeline owner's move, but the landowner does not observe either, resulting in a simultaneous move game between the state and the landowner. As a result, the interaction between the state and the landowner is modeled as a simultaneous move game, analyzed under two scenarios—one where the pipeline owner complies, and one where they do not. Because the landowner does not know which node they occupy, each condition generates a distinct subgame requiring separate analysis at the final stage. When the pipeline owner chooses action a (compliance), the resulting normal form game between the state and the landowner is:

State / Landowner	e	f
c	<u>740, 87</u>	<u>752, 79</u>
d	550, <u>86</u>	580, 76

where the Nash equilibrium is at strategy pair (c,e), yielding a payoff of (740,87) to the state and the landowner, respectively. Alternatively, if the pipeline owner chooses action b (cost-cutting), the normal form game between the state and the landowner becomes:

State / Landowner	e	f
c	<u>640, 81</u>	<u>670, 68</u>
d	400, <u>81</u>	450, 67

where the Nash equilibrium is at strategy pair (c,e), yielding a payoff of (640,81) to the state and the landowner, respectively.

Given the equilibrium payoffs, the pipeline owner compares outcomes and finds that compliance yields a higher return: 5184 under the strategy profile (a, c, e), compared to 1600 under (b, c, e). Anticipating this, the pipeline owner rationally chooses compliance, completing the backward induction and resulting in a subgame perfect Nash equilibrium where the state monitors consistently and the landowner takes precaution (a, c, e). This equilibrium reflects a coordination game shaped by endogenous risk and imperfect information, where cautious behavior arises not from mutual distrust or external enforcement, but from each player's incentive to reduce exposure to collective risk. Even without full transparency, decentralized actors align on safer strategies when the cost of negligence is internalized. In practice, mechanisms such as fines, liability funds, or adjusted insurance premiums that increase the direct cost of negligence may encourage voluntary compliance and reinforce the self-regulating dynamics reflected in the model.

Extension Prospects

This framework can be extended through scenario simulations, empirical calibration, and sensitivity analysis to examine how variations in leak probability, compliance cost, and environmental damage affect equilibrium outcomes and payoffs. Future studies may incorporate deeper forms of incomplete information such as Bayesian games or dynamic programming models to capture belief formation, strategic uncertainty, and intertemporal decision making, or extend the analysis to compare decentralized outcomes with socially optimal allocations through a planner's optimization framework.

Conclusion

This study presents a game theoretical framework in which the probability of a pipeline leak and the distribution of environmental damages are endogenously shaped by the strategic decisions of the pipeline owner, the regulator, and the landowner. By tying damage shares to each actor's level of care and introducing imperfect information, the model shows how internalized incentives, rather than external enforcement, can lead to voluntary coordination on responsible behavior, particularly when the cost of damage is high enough compared to the cost of taking precautionary measures on individual levels.

References

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Appendix

The utility expressions for each player are derived by substituting the weight values for each strategy combination from the game tree as below:

$$(1) \text{ a,c,e: } P = \frac{1}{10000} \left(\pi - C_P - \frac{0.2}{3} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{0.2}{3} M \right), L = \left(\sqrt{\lambda - C_L - \frac{0.2}{3} qM} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{1}{15} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{1}{15} M \right), L = \left(\sqrt{\lambda - C_L - \frac{1}{15} qM} \right)$$

$$(2) \text{ a,c,f: } P = \frac{1}{10000} \left(\pi - C_P - \frac{0.25}{4} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{0.25}{4} M \right), L = \left(\sqrt{\lambda - \frac{0.25}{2} M} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{1}{16} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{1}{16} M \right), L = \left(\sqrt{\lambda - C_L - \frac{1}{8} qM} \right)$$

$$(3) \text{ a,d,e: } P = \frac{1}{10000} \left(\pi - C_P - \frac{0.3}{4} M \right)^2, S = \frac{1}{10} \left(\sigma - \frac{0.3}{2} M \right), L = \left(\sqrt{\lambda - C_L - \frac{0.3}{4} M} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{3}{40} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{3}{20} M \right), L = \left(\sqrt{\lambda - C_L - \frac{3}{40} qM} \right)$$

$$(4) \text{ a,d,f: } P = \frac{1}{10000} \left(\pi - C_P - \frac{0.35}{5} M \right)^2, S = \frac{1}{10} \left(\sigma - \frac{2*0.35}{5} M \right), L = \left(\sqrt{\lambda - \frac{2*0.35}{5} M} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{7}{100} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{7}{50} M \right), L = \left(\sqrt{\lambda - C_L - \frac{7}{50} qM} \right)$$

$$(5) \text{ b,c,e: } P = \frac{1}{10000} \left(\pi - \frac{0.4}{2} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{0.4}{4} M \right), L = \left(\sqrt{\lambda - C_L - \frac{0.4}{4} M} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{1}{5} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{1}{10} M \right), L = \left(\sqrt{\lambda - C_L - \frac{1}{10} qM} \right)$$

$$(6) \text{ b,c,f: } P = \frac{1}{10000} \left(\pi - \frac{2*0.45}{5} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{0.45}{5} M \right), L = \left(\sqrt{\lambda - \frac{2*0.45}{5} M} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{9}{50} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{9}{100} M \right), L = \left(\sqrt{\lambda - C_L - \frac{9}{50} q M} \right)$$

$$(7) \text{ b,d,e: } P = \frac{1}{10000} \left(\pi - \frac{2*0.5}{5} M \right)^2, S = \frac{1}{10} \left(\sigma - \frac{2*0.5}{5} M \right), L = \left(\sqrt{\lambda - C_L - \frac{0.5}{5} M} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{1}{5} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{1}{5} M \right), L = \left(\sqrt{\lambda - C_L - \frac{1}{10} q M} \right)$$

$$(8) \text{ b,d,f: } P = \frac{1}{10000} \left(\pi - \frac{0.55}{3} M \right)^2, S = \frac{1}{10} \left(\sigma - \frac{0.55}{3} M \right), L = \left(\sqrt{\lambda - \frac{0.55}{3} M} \right)$$

$$\Rightarrow \frac{1}{10000} \left(\pi - C_P - \frac{11}{60} M \right)^2, S = \frac{1}{10} \left(\sigma - C_S - \frac{11}{60} M \right), L = \left(\sqrt{\lambda - C_L - \frac{11}{60} q M} \right)$$

The payoffs from the utility functions using the values $\pi = \sigma = \lambda = 10,000$; $C_P = 800$; $C_S = 600$; $C_L = 400$; $M = 30,000$ that are plotted in the game tree were calculated using the following R code:

```
# Parameters
pi <- 10000
sigma <- 10000
lambda <- 10000
CP <- 800
CS <- 600
CL <- 400
M <- 30000

# Function to calculate q based on actions
get_q <- function(prod_action, state_action, land_action) {
  base_q <- 0.2
  q <- base_q +
    ifelse(prod_action == "b", 0.2, 0) +
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    ifelse(state_action == "d", 0.1, 0) +
    ifelse(land_action == "f", 0.05, 0)
  return(q)}

# Function to calculate weights
get_W <- function(prod_action, state_action, land_action) {
  WP <- ifelse(prod_action == "a", 1, 2)
  WS <- ifelse(state_action == "c", 1, 2)
  WL <- ifelse(land_action == "e", 1, 2)
  W <- WP + WS + WL
  return(list(WP=WP, WS=WS, WL=WL, W=W))}

# Function to calculate payoffs
calculate_payoffs <- function(prod_action, state_action, land_action) {
  q <- get_q(prod_action, state_action, land_action)
  weights <- get_W(prod_action, state_action, land_action)
  prod_damage <- q * M * (weights$WP / weights$W)
  state_damage <- q * M * (weights$WS / weights$W)
  land_damage <- q * M * (weights$WL / weights$W)
  P <- if (prod_action == "a") {
    ((pi - CP - prod_damage)^2) / 10000
  } else {
    ((pi - prod_damage)^2) / 10000
  }
  S <- if (state_action == "c") {
    (sigma - CS - state_damage) / 10
  } else {
    (sigma - state_damage) / 10
  }
  L <- if (land_action == "e") {
    sqrt(lambda - CL - land_damage)
  } else {

```

```

    sqrt(lambda - land_damage)
  }
  return(c(P=round(P), S=round(S), L=round(L)))}

# Define strategy profiles
strategies <- list(
  ace = c("a", "c", "e"),
  acf = c("a", "c", "f"),
  ade = c("a", "d", "e"),
  adf = c("a", "d", "f"),
  bce = c("b", "c", "e"),
  bcf = c("b", "c", "f"),
  bde = c("b", "d", "e"),
  bdf = c("b", "d", "f"))

# Calculate and store payoffs
results <- data.frame(
  Strategy = character(),
  Producer = numeric(),
  State = numeric(),
  Landowner = numeric())
for (name in names(strategies)) {
  actions <- strategies[[name]]
  payoffs <- calculate_payoffs(actions[1], actions[2], actions[3])
  results <- rbind(results, data.frame(
    Strategy = name,
    Producer = payoffs[1],
    State = payoffs[2],
    Landowner = payoffs[3]))}

```